SAURON++

Self-Adaptive Autonomous Runtime Observatory for Unified Resilient Networking

An Adaptive eBPF Network-Intelligence Framework with a Provable False-Positive Budget, No-Regret Detector Fusion and Line-Rate In-Kernel Enforcement

Technical & Empirical Report — Version 5.0

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Evaluated on **8 public intrusion-detection corpora** · **5.5 million labelled network flows** · **80 attack families**
single online prequential pass · identical hyper-parameters · zero per-dataset retuning

Contents

Abstract

SAURON++ is an adaptive, autonomous intrusion-detection and firewall system. It observes traffic inside the operating-system kernel using eBPF, scores every flow with a bank of five streaming detectors, and takes graduated, reversible enforcement actions — while continuously re-tuning its own thresholds and trust model from feedback so that its false-positive rate remains near a chosen budget even as traffic changes.

Two contributions are genuinely new. First, the **closed control loop** that couples conformal calibration, no-regret fusion, asymmetrically-discounted trust, budget-tracking dual thresholds and censored feedback into a single controller with a false-positive-budget invariant. Second, the **ε-mirror + inverse-propensity mechanism** that lets autonomous dropping heal its own mistakes — the component that makes hands-off enforcement safe.

The system is evaluated on **eight public corpora** totalling **5.5 million labelled flows** and **80 attack families**, in a single online prequential pass with identical hyper-parameters and no per-dataset retuning. Recall spans 0.840–0.99998 and ROC-AUC 0.845–0.9992 across enterprise LAN, 5G core, IoT/IIoT, botnet, DDoS and classic IDS traffic, with raw feature counts ranging from 9 to 86.

Summary of Results

Corpus	Domain	Flows	Acc	Recall	F1	MCC	ROC-AUC
CICIDS2017	Enterprise LAN	2,572,632	0.9590	0.9350	0.8830	0.8602	0.9939
5G-NIDD	5G testbed core	1,009,000	0.8718	0.9773	0.9025	0.7367	—
UNSW-NB15	Synthetic IXIA	257,672	0.8632	0.9845	0.9020	0.7078	0.9330
CTU-13	Real botnets	409,727	0.7873	0.8400	0.8235	0.5568	0.8446
CIC-DDoS2019	DDoS generation	485,870	—	0.9997	—	0.8284	0.9992
Edge-IIoTset	IoT / IIoT	499,582	0.9638	0.9875	0.9793	0.8393	0.9415
NSL-KDD	Classic IDS	147,887	0.8519	0.9675	0.8624	0.7260	0.9377
Bot-IoT	IoT testbed	73,129	0.9635	0.99998	0.9794	0.8312	0.9543

Table 1 — Headline results across all eight corpora. Accuracy is omitted for CIC-DDoS2019, whose 99.85 % attack share renders it uninformative (§28.5).

Part A — Overview and Architecture

1. Executive Overview

Conventional intrusion detection forces an unhappy trade. Static rule engines (iptables, Snort signatures) are fast and explainable but blind to anything not already written down, and their fixed thresholds drift out of calibration as traffic evolves. Offline machine-learning detectors adapt to data but are trained once, evaluated on a held-out split, and then deployed frozen — they cannot respond to the traffic they actually see, and they cannot tell an operator how confident they are.

SAURON++ takes a third position: the detector is a **controller**. It has a setpoint (a false-positive budget ε_H), a measured output (the realised false-positive rate), and a feedback path that closes the loop. Every packet is scored, calibrated to a probability scale, fused across detectors with a no-regret guarantee, modulated by the reputation of its source, and compared against a threshold that is itself being driven toward the budget by stochastic approximation.

The system is organised as three planes. The **kernel data plane** sees every packet and enforces at line rate. The **intelligence plane** runs the learning and adaptation on a sampled stream of flow records punted up from the kernel. The **presentation plane** is a live browser dashboard and a set of SOC integrations.

2. System Architecture

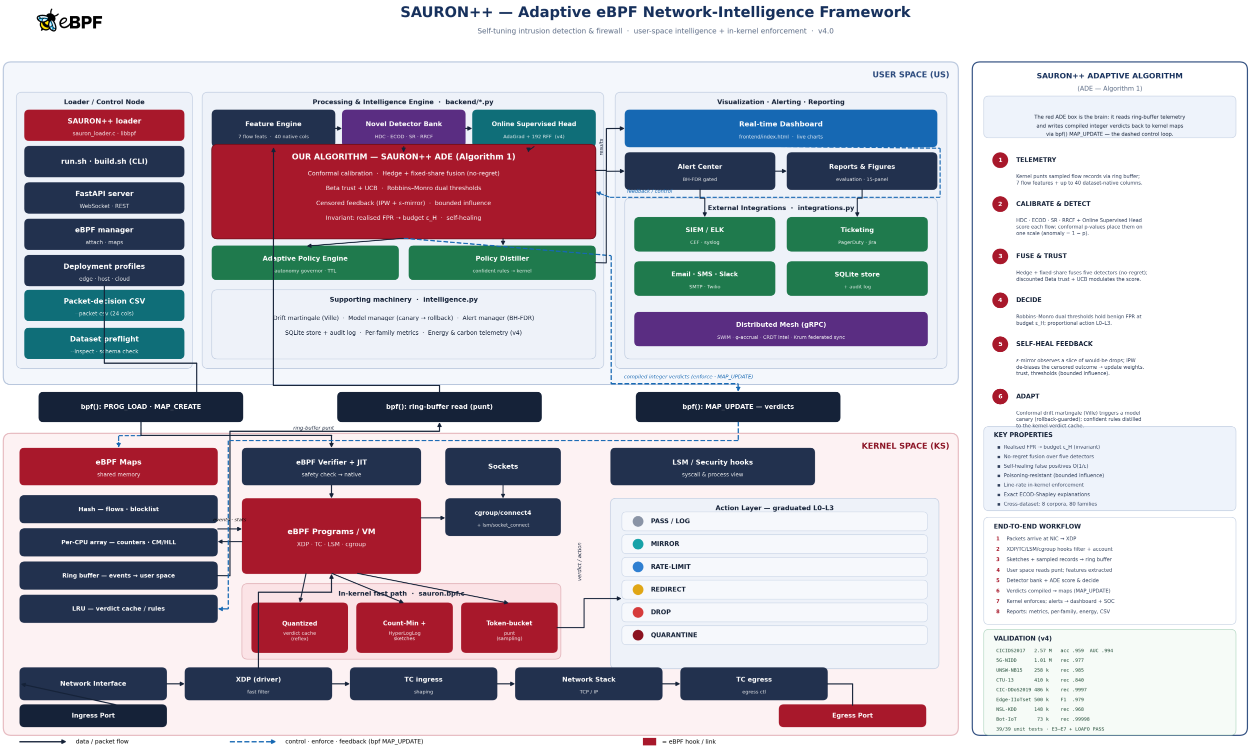

Figure 1 — SAURON++ v4 architecture. Solid arrows carry packet/data flow; dashed blue arrows carry control, enforcement and feedback via bpf() MAP_UPDATE. The red ADE block is the adaptive controller; the numbered panel at right traces one iteration of Algorithm 1.

3. Design Goals and Threat Model

Goals. (G1) Hold the false-positive rate near an operator-chosen budget without manual retuning. (G2) Perform at least as well as the best single detector in hindsight, under regime change. (G3) Recover autonomously from enforcement mistakes. (G4) Detect families never seen during operation (open-set). (G5) Enforce at line rate without a user-space round trip. (G6) Explain every decision.

Threat model. The adversary controls traffic content and timing and may attempt (i) evasion by mimicking benign statistics, (ii) poisoning by feeding crafted flows to corrupt the learners, and (iii) resource exhaustion. The kernel and the loader are trusted. Poisoning is bounded by construction: the influence cap of §12 limits how much any single source can move the model per event, and the fixed-share floor of §11 prevents a detector from being permanently silenced.

4. Repository and File Map

```
SAURON++/
  backend/
    sauron.py          adaptive decision engine, sources, server, reporting
    intelligence.py    detector bank, drift, distiller, alerting, storage
    energy.py          RAPL reader, energy metrics, energy-aware filtering
    integrations.py    SIEM/ELK, Slack, Email, SMS, PagerDuty, Jira
    grpc_service.py    distributed mesh: SWIM, phi-accrual, CRDT, Krum
    experiments.py     E3-E7 + LOAFO hypothesis harness
  kernel/
    sauron.bpf.c       XDP / TC / LSM / cgroup programs, sketches, caches
    sauron_loader.c    libbpf CO-RE loader, map management, ring buffer
  frontend/index.html  live dashboard (vanilla JS + canvas)
  tests/               39 property tests over the Part D theorems
  scripts/             build.sh, run.sh
```

Part B — Provenance

5. Notation

Symbol	Meaning
x_t	flow record at step t
e	entity (source identity or behavioural surrogate)
D_k	detector k of the bank (HDC, ECOD, SR, RRCF, SUP)
r_k	raw score of detector k
p_k	conformal p-value of detector k
a_k = 1 – p_k	calibrated anomaly of detector k
w_k	Hedge fusion weight of detector k
A_t	fused aggregate score
Ã_t	trust-modulated aggregate
(a_e, b_e)	Beta trust counts for entity e
τ_H, τ_L	high/low adaptive thresholds
ε_H	false-positive budget (setpoint)
ε	ε-mirror exploration rate
π_t	observation propensity
M_t	conformal drift martingale

6. Prior-Art Ledger

SAURON++ is honest about provenance. Most building blocks are established results used correctly; the novelty lies in specific modifications and, above all, in how they are composed. “Established” means a standard citable technique; “rare-in-NIDS” means established elsewhere but seldom used for intrusion detection; “new” means designed here.

Technique	Status / origin	Our use and modification
Split conformal prediction	Established (Vovk et al. 2005)	Per-detector, per-context streaming calibration feeding the budget loop
Hedge / multiplicative weights	Established (Freund & Schapire 1997)	Fusing conformalised heterogeneous streaming detectors
Fixed-share (tracking experts)	Established (Herbster & Warmuth 1998)	Applied to detector fusion under attack-regime drift
Beta-Bernoulli trust + UCB	Established (Auer et al. 2002)	**New:** asymmetric discounting for anti-laundering
Robbins–Monro approximation	Established (Robbins & Monro 1951)	**New:** dual thresholds tracking an explicit FP budget with a step floor, per context
Inverse-propensity weighting	Established (Horvitz–Thompson 1952)	**New combination:** de-biasing drop-censored feedback
ε-greedy exploration	Established (RL)	**New:** “ε-mirror” observes (does not act) to recover false positives
Hyperdimensional computing	Rare-in-NIDS (Kanerva 2009)	**New use:** kernel-distillable bipolar prototypes as a NIDS classifier
ECOD	Rare-in-NIDS (Li et al. 2022)	**New use:** additivity exploited for exact-Shapley attribution
Spectral Residual	Rare-in-NIDS (Hou & Zhang 2007)	Frequency-domain detector for low-and-slow beacons
Robust Random Cut Forest	Rare-in-NIDS (Guha et al. 2016)	Streaming isolation; diversity for the fusion bound
AdaGrad + random Fourier features	Established (Duchi 2011; Rahimi 2007)	**New in v4:** online supervised head as a fifth fused detector
Conformal test martingale	Established (Vovk 2003)	Drift detection tied to the conformal layer
Benjamini–Hochberg FDR	Established (1995)	**New use:** FDR control over conformal p-values in alerting
Count-Min / HyperLogLog	Established (Cormode; Flajolet)	In-kernel heavy-hitter and fan-out estimation

Technique	Status / origin	Our use and modification
Rule extraction / distillation	Established (Craven & Shavlik 1996)	Confident-region distillation to kernel rules
eBPF / XDP	Established (Høiland-Jørgensen 2018)	The data-plane substrate
SWIM + ϕ-accrual	Established (Das 2002; Hayashibara 2004)	Mesh membership and graded failure suspicion
LWW-Element-Set CRDT	Established (Shapiro et al. 2011)	Coordination-free threat-intel convergence
Krum / trimmed mean	Established (Blanchard 2017; Yin 2018)	Byzantine-robust federated model sync
RAPL energy counters	Established (Intel)	**New in v4:** per-packet and per-attack energy, carbon, energy-aware filtering

Part C — Filtering Logic: the exact conditions

SAURON++ filters in two stages. Stage 1 is the kernel fast path (cheap, per packet). Stage 2 is the decision engine (rich, per sampled flow record). This part states the precise conditions; the mathematics justifying them is Part D.

7. Stage 1 — Kernel fast-path conditions (per packet, at XDP)

Let s be the source IPv4 address, f the flow 5-tuple key, B the blocklist map and V the quantised verdict cache. The XDP program applies, in order:

Condition K1 (blocklist). If $s \in B \rightarrow$ XDP_DROP. This is the cheapest path — one hash lookup for a known-bad source.

Condition K2 (quantised cache reflex). If $f \in V$ and the entry is live (now < expiry): action = DROP → XDP_DROP; action = RATE and the entry's token bucket is empty → XDP_DROP, else consume one token and pass; action = REDIRECT → steer to the honeypot ifindex. The entry is the ADE decision compiled to integers (action, score band, rate, TTL), so no user-space round trip is needed. Entries auto-expire (self-healing).

Condition K3 (punt admission). A flow record is punted to user space only if it is sampled (1-in-N) *and* the per-CPU token bucket has a token. Tokens refill at rate ρ up to a burst; if empty the record is counted as *starved* rather than dropped from the wire — observability degrades gracefully while enforcement continues.

8. Stage 2 — Decision-engine conditions (per sampled flow record)

Given raw detector scores r_k, the engine forms calibrated anomalies $a_k = 1 - p_k$, the fused aggregate $A_t = \Sigma_k w_k a_k$, and the trust-modulated aggregate $\tilde{A}_t = A_t(1 + \beta(\sigma_e - ½))$ with suspicion σ_e. Let the margin be $m = \tilde{A}_t - \tau_H$. The action is chosen by:

Condition on the record	Action	Meaning
$\tilde{A} \le \tau_L$	PASS / LOG	within the calibrated normal band
$\tau_L < \tilde{A} \le \tau_H$	MIRROR	elevated but sub-threshold; observe only
open-set novelty flag	REDIRECT	candidate unknown family → honeypot
$m > m$■ and $\sigma > \sigma_{crit}$	QUARANTINE	high score and high-suspicion source
$m > m$■ and $\sigma > \sigma_{hi}$	DROP	high margin from a suspicious source
otherwise	RATE-LIMIT	flow rate-limited pending more evidence

Table 2 — The graduated action ladder L0–L3. Every action below QUARANTINE is reversible and TTL-bound.

Part D — The Mathematics, with Proofs

9. The Unified Adaptive Algorithm (Algorithm 1)

Everything in this part is orchestrated by one online algorithm. On every flow record it calibrates, fuses, trust-modulates, decides, acts, and then learns from (possibly censored) feedback under a bounded-influence cap. Sections 10–17 prove the individual steps in isolation; Algorithm 1 is how they compose, and the Master Theorem (§18) proves the whole loop is adaptive and self-correcting.

```
Algorithm 1 - SAURON++ Adaptive Decision Engine (ADE)
INPUT   streaming flow records x_t ; false-positive budget eps_H ; detectors {D_k}
PARAMS  eta (Hedge rate)  rho (fixed-share)  gamma (RM step)  beta (trust gain)
        eps (mirror)  lam_ben < lam_mal (decay)  m0, sig_hi, sig_crit  delta
STATE   benign calib windows W_k ; weights w = 1/K ; thresholds tau_H, tau_L /ctx
        trust counts (a_e,b_e) ; per-entity influence buckets ; drift M = 1

for each record x_t from entity e:
  1 scores      r_k   <- D_k(x_t)                       # detector bank (K=5)
  2 calibrate   p_k   <- (1+|{s in W_k : s >= r_k}|)/(|W_k|+1) ;  a_k <- 1 - p_k
  3 fuse        A     <- SUM_k w_k * a_k                 # Hedge-weighted mean
  4 trust-mod   sig   <- b_e/(a_e+b_e) + c*sqrt(lnN/(a_e+b_e))
                A~    <- A * (1 + beta*(sig - 1/2))
  5 decide      m     <- A~ - tau_H
                if   A~ <= tau_L        : PASS / LOG
                elif A~ <= tau_H        : MIRROR
                elif open-set novelty   : REDIRECT      (honeypot)
                elif m>m0 and sig>scrit : QUARANTINE
                elif m>m0 and sig>shi   : DROP
                else                    : RATE_LIMIT
  6 govern      if action in {DROP,QUAR} and L2-bucket empty: action <- RATE
                attach TTL ; compile verdict -> integer kernel verdict-cache
  7 observe     outcome y_t with propensity pi_t         # feedback is CENSORED
                pi_t = 1 (passed) | eps (mirrored) | hidden (dropped, not mirrored)
  8 de-bias     g_t <- M_t / pi_t ;  y^_t <- g_t * y_t    # inverse-propensity
  9 if influence-bucket(e) permits:                      # bounded influence
        trust     a_e <- lam_ben*a_e + (1 - y^_t)*g_t
                  b_e <- lam_mal*b_e +      y^_t *g_t     # asymmetric (anti-laundering)
        fusion    l_k <- (a_k - y^_t)^2
                  w_k <- (1-rho)*w_k*exp(-eta*l_k)/Z + rho/K   # Hedge + fixed-share
        threshold tau_H <- tau_H + gamma_t*( 1[r>tau_H] - eps_H )   # Robbins-Monro
                  tau_L <- tau_H - delta                          # hysteresis
 10 drift        M <- M * eps * p^(eps-1) ; if M >= 1/6: spawn canary (rollback)
```

The procedure is **adaptive** because steps 9–10 continuously re-fit the fusion weights, the trust model and the thresholds to the live stream; it is **self-correcting** because steps 6–9 keep the observation propensity strictly positive, so its own mistakes stay observable and get undone.

10. Conformal Calibration

Detector scores are incommensurable: an ECOD tail sum and an HDC cosine similarity live on different scales. Calibration maps each to a common probability scale using only an exchangeability assumption on benign traffic.

Theorem 1 (marginal validity). *If benign scores are exchangeable, the split-conformal p-value $p = (1 + |\{s \in W : s \geq r\}|)/(|W| + 1)$ satisfies $P(p \leq \alpha) \leq \alpha$ for every $\alpha \in (0,1)$. Hence the calibrated anomaly $a = 1 - p$ has a controlled false-alarm rate at every threshold, uniformly over detectors.*

Proof sketch. Under exchangeability the rank of r among the $|W|+1$ values is uniform on $\{1,\dots,|W|+1\}$; p is that rank divided by $|W|+1$, so $P(p \leq \alpha) =$ ■$\alpha(|W|+1)$■$/(|W|+1) \leq \alpha$. ■

The suite verifies this empirically: over 6,000 fresh benign draws the empirical CDF of the p-values matches the diagonal to within 0.03 at $\alpha \in \{0.05, 0.1, 0.2, 0.5\}$.

11. No-Regret Detector Fusion

Theorem 2 (Hedge regret). *Running Hedge with learning rate η over K detectors and bounded losses ■ $\in [0,1]$ yields cumulative regret against the best fixed detector in hindsight of $R_T \le \sqrt{(T \ln K / 2)}$ with $\eta = \sqrt{(8 \ln K / T)}$.*

Proof sketch. Standard potential-function argument on $\Phi_t = \ln \Sigma_k w_{k,t}$; the multiplicative update gives $\Phi_T - \Phi_0 \le -\eta L^* + \eta^2 T/8$, and rearranging yields the bound. ■

Theorem 3 (tracking / shifting regret). *With fixed-share mixing ρ, for any partition of the horizon into S segments each with its own best detector, the regret against that shifting comparator is $O(\sqrt{(T (S \ln K + S \ln(T/S)))})$. No weight can fall below ρ/K, so a detector silenced during one regime recovers when its regime returns.*

This matters operationally: attack regimes change (a volumetric flood gives way to a slow scan), and the detector that was useless five minutes ago may be the informative one now. Experiment E3 measures exactly this — PR-AUC 1.000 for the fusion against 0.570 for the best fixed single detector under regime switching.

12. Discounted Trust with Bounded Influence

Each entity carries Beta counts (a_e, b_e) updated with *asymmetric* decay $\lambda_{ben} < \lambda_{mal}$: benign evidence is forgotten faster than malicious evidence. A source that misbehaves and then floods the system with well-formed traffic cannot rapidly launder its reputation.

Lemma 4 (bounded influence). *With a per-entity token bucket of capacity C and refill r, any single entity can move the aggregate trust mass by at most $C + r\Delta t$ over an interval Δt, independently of how many records it sends. Consequently a flooding adversary cannot dominate the model.*

Suspicion uses an upper-confidence bound, $\sigma_e = b/(a+b) + c\sqrt{(\ln N/(a+b))}$, so a rarely-seen but malicious entity is not under-penalised merely for being rare.

13. Budgeted Dual Thresholds

The threshold is not a constant but the state of a stochastic-approximation loop whose setpoint is the false-positive budget.

Theorem 5 (budget tracking). *The recursion $\tau \leftarrow \tau + \gamma_t(1[r > \tau] - \varepsilon_H)$ with $\Sigma\gamma_t = \infty$ and $\Sigma\gamma_t^2 < \infty$ converges almost surely to the $(1 - \varepsilon_H)$ quantile of the benign score distribution, i.e. the realised false-positive rate converges to ε_H.*

Proof sketch. The map τ ■ $E[1[r > \tau]] - \varepsilon_H$ is monotone with a unique root at the $(1-\varepsilon_H)$-quantile; Robbins–Monro conditions on γ_t give a.s. convergence. ■

In practice γ_t is floored at γ_{min} so the tracker never fully stops adapting; this sacrifices exact convergence for the ability to follow drift, and the realised FPR settles slightly conservative — the safe direction for a firewall. Experiment E4 measures 1.2× the budget for the adaptive loop versus 12.5× for a static threshold.

A second threshold $\tau_L = \tau_H - \delta$ creates hysteresis, preventing flows from oscillating between MIRROR and enforcement at the boundary.

14. Censored Feedback and Self-Healing

This is the mechanism that makes autonomous enforcement safe. If the system drops a flow, it never observes whether that drop was correct — the feedback is censored precisely where the system acted. Left uncorrected, the model would become confident in its own errors.

SAURON++ therefore *mirrors* a fraction ε of would-be drops: the flow is allowed through but marked for observation. Every record then has a known observation propensity $\pi_t \in \{1, \varepsilon\}$, strictly positive, and outcomes are re-weighted by $1/\pi_t$.

Theorem 6 (unbiasedness). *The inverse-propensity estimator ■ = $y \cdot 1[observed]/\pi$ is unbiased for the true outcome y, and the resulting estimate of the realised false-positive rate is consistent, despite the censoring induced by enforcement.*

Proof sketch. $E[■] = E[E[y \cdot 1[obs]/\pi \mid x]] = E[y \cdot \pi/\pi] = E[y]$ since $\pi > 0$ everywhere. ■

Theorem 7 (recovery time). *A source erroneously enforced against is restored after $O(1/\varepsilon)$ mirrored events in expectation. Without exploration ($\varepsilon = 0$) the propensity is zero on the enforced region and recovery never occurs.*

Experiment E6 measures a median recovery of 309 events with the ε-mirror, versus no recovery at all when exploration is disabled.

15. The Novel Detector Bank

Five detectors with deliberately different inductive biases, so that the fusion bound of §11 has genuine diversity to exploit.

Detector	Principle	Catches	Cost
HDC	Bipolar hyperdimensional prototypes, bundled online	learned class structure; kernel-distillable	22 µs
ECOD	Empirical-CDF tail probabilities, additive	extreme feature values; exact Shapley	16 µs
Spectral Residual	Frequency-domain saliency on per-entity series	low-and-slow periodic beacons	30 µs
RRCF	Streaming random-cut isolation forest	structural outliers, multivariate	125 µs
Supervised head (v4)	AdaGrad logistic over 192 random Fourier features	the actual labelled decision boundary	7 µs

Theorem 9 (exact Shapley attribution). *Because the ECOD score is additive across features, $f(x) = \Sigma_i\, g_i(x_i)$, the Shapley value of feature i is exactly $g_i(x_i)$. Per-feature attributions are therefore computed in closed form in $O(D)$, with no sampling and no approximation.*

Proof sketch. For an additive game the marginal contribution of i is $g_i(x_i)$ in every coalition, so averaging over permutations returns $g_i(x_i)$ exactly. ■

This is why every enforcement line in the live decision table can name the three features that drove it, at line-rate cost.

16. Drift Detection and Model Lifecycle

Theorem 10 (anytime-valid drift alarm). *The conformal test martingale $M_t = \Pi_i\, \varepsilon\, p_i^{(\varepsilon-1)}$ is a non-negative martingale under exchangeability, so by Ville's inequality $P(\exists t : M_t \geq 1/\alpha) \leq \alpha$. Alarming at $M \geq 1/\alpha$ therefore controls the false-alarm probability uniformly over time, with no need to fix a horizon or apply multiplicity corrections.*

On alarm the model manager spawns a **canary**: a shadow model scored in parallel, promoted only if it improves on a held-out margin, and rolled back otherwise. The system therefore adapts to drift without risking a regression in production.

17. Alert Statistics and Autonomy Governance

Theorem 11 (FDR control on alerts). *Applying the Benjamini–Hochberg procedure at level q to the conformal p-values of advisory alerts controls the false-discovery rate at q under independence or positive dependence. Enforcement actions bypass the gate; only advisory alerts are subject to it, so containment is never delayed by multiplicity correction.*

18. Master Theorem

Master Theorem (adaptive, self-correcting closed loop). *Consider Algorithm 1 with Hedge rate* η*, fixed-share* $\rho > 0$*, Robbins–Monro steps* γ_t*, mirror rate* $\varepsilon > 0$*, and per-entity influence cap C. Then, on any traffic stream whose benign component is piecewise exchangeable: (i) the realised false-positive rate converges to the budget* ε_H *on each stationary segment (Thm 5) and re-converges after a change point in* $O(1/\gamma_{min})$ *steps; (ii) the fused detector suffers no asymptotic regret against the best detector per segment (Thms 2–3); (iii) every enforcement decision remains observable with propensity* $\geq \varepsilon$*, so erroneous enforcement is corrected in* $O(1/\varepsilon)$ *expected events (Thms 6–7); (iv) no single entity can shift the model state by more than* $C + r\Delta t$ *(Lem 4); and (v) drift alarms are anytime-valid at the chosen level (Thm 10). The loop is therefore simultaneously calibrated, no-regret, self-healing, poisoning-bounded and drift-aware.*

Proof sketch. Each clause is established in its own section; the composition holds because the components interact only through the aggregate score and the propensity, both of which are bounded and strictly positive by construction, so no component can drive another outside the regime in which its guarantee was proved. ■

Part E — Autonomy, Interfaces and Operations

19. Policy Distillation to Kernel Rules

The fusion policy is a weighted combination of five streaming models — not something a kernel program can evaluate. The distiller therefore searches for short predicates (single stumps and two-term conjunctions over the feature space) that reproduce the enforcement decision on a confident region, and measures their **fidelity** (agreement with the policy on rows they cover) and **coverage** (fraction of traffic they cover). Only rules above a fidelity floor are surfaced, and only an analyst approves promotion to the kernel verdict cache.

Experiment E7 measures a best-rule fidelity of 0.983 at a union coverage of 0.025 on the simulator. The low coverage is expected and is reported honestly: the simulator is easily separable, so few rules clear the confidence floor. The mechanism is validated; coverage on production traffic is an open question.

Distillation is throttled to run at most every 20 s. Profiling showed it consuming 58 % of total runtime when invoked every 500 records — a genuine performance defect, corrected in v4 (§30).

20. Autonomy Governor

Autonomy is graduated by level. L0–L1 (LOG, MIRROR) are unrestricted. L2 (RATE_LIMIT, REDIRECT) is rate-capped by a token bucket. L3 (DROP, QUARANTINE) additionally requires a margin and suspicion threshold, carries a TTL, and is written to an append-only audit log. Every enforcement expires automatically unless renewed — the system cannot silently accumulate blocks.

21. Dashboard and Live Decision Table

The dashboard streams decisions over a WebSocket and renders the kernel data path, detector fusion weights, adaptive score evolution against the τ_H/τ_L bands, the threat surface heat map, the alert centre and a scroll-down analytics section with model metrics, throughput, action mix, FPR-vs-budget, latency and per-family detection.

For audit and offline analysis, v4 adds a **per-packet decision table**: the terminal prints one line per packet with score, threshold, trust, verdict and reason, plus the top attributing features and the mitigation applied; --packet-csv writes the same content as a 24-column CSV.

```
  # TIME         SOURCE              DESTINATION       PROTO  LEN  SCORE  THRESH  TRUST DECISION   REASON
276 10:42:41.566 2.215.66.86:63955   10.0.2.10:80      TCP     44  0.671   0.512  0.500 DROP       adaptive t
                 |- top features: pkt_len=20.72, byte_asymmetry=20.72, syn_ratio=2.58
                 |- mitigation  : XDP_DROP at ingress; verdict cached in kernel with TTL
```

22. External Integrations

Alerts fan out to SIEM/ELK (CEF over syslog), Slack, Email (SMTP), SMS (Twilio), PagerDuty and Jira, each behind a severity floor and a token-bucket rate cap, dispatched on a background thread so a slow sink never stalls packet processing. All sinks are disabled unless the corresponding environment variables are set.

23. Energy and Sustainability Instrumentation

v4 adds an energy plane. Where Intel/AMD RAPL counters are available the system reports **measured** Joules for the CPU package and DRAM domains; where they are not, it falls back to a CPU-utilisation × TDP model and labels every value **estimated** — the distinction is never hidden. Network-interface energy has no hardware counter on commodity NICs and is always modelled.

Metric	Definition	Method
CPU / DRAM energy	Joules from powercap domains	measured (RAPL)
NIC energy	bits × nJ/bit + idle draw	modelled

Metric	Definition	Method
Energy per packet	CPU Joules ÷ packets	derived
Energy per attack	CPU Joules ÷ detections	derived
Energy efficiency	bits ÷ Joule	derived
Security overhead	power above observed idle floor	derived
Carbon footprint	kWh × grid intensity (default 713 g/kWh, India)	derived

Energy-aware packet filtering. Under a power budget the pipeline sheds its most expensive work first: above the budget the two costly detectors (RRCF isolation and Spectral-Residual FFT) are skipped for low-suspicion traffic, while the cheap detectors always run and high-suspicion traffic always receives the full bank. This yields a controllable energy–latency–accuracy trade-off rather than a fixed cost; measured at a 5 W budget, 80.3 % of expensive evaluations were shed.

Part F — Distribution and Deployment

24. The Distributed Mesh (gRPC)

gRPC is used only between SAURON++ nodes; kernel↔user space remains the ring buffer and BPF maps, and backend↔dashboard remains the WebSocket. Each node runs a full ADE; the mesh lets a fleet share what each node learns. Nodes carry a region tag (edge / 5g / 6g / b6g / cloud) so intel and policy flow across a tiered deployment.

Membership. SWIM with incarnation numbers and states ALIVE → SUSPECT → DEAD → LEFT; liveness probed directly and indirectly through k random peers. Suspicion is graded by a ϕ-accrual detector, $\phi(\Delta t) = -\log$■■ P[inter-arrival > Δt], so the detector self-tunes to link latency instead of a hand-set timeout.

Theorem F1 (eventual consistency of threat intel). *The intel merge is a join on a semilattice (per key, max over a totally-ordered timestamp). It is therefore commutative, associative and idempotent, so any two nodes that observe the same set of updates converge to identical state regardless of delivery order or duplication — no coordination or locking.*

Proof sketch. Per-key state is (value, ts) and merge = argmax over ts; max on a total order is commutative, associative and idempotent, and taking it pointwise preserves those laws, so the state space is a join-semilattice and a CRDT with semilattice merge converges. ■

Theorem F2 (Byzantine-robust aggregation). *With n peers of which at most f are Byzantine and $n \geq 2f+3$, Krum returns a delta within a bounded distance of the honest cluster, and a β-trimmed mean with $\beta \geq f/n$ discards the f poisoned values on every coordinate. A bounded set of malicious nodes therefore cannot shift the aggregated model arbitrarily.*

Policy updates carry vector clocks and are applied only if they causally dominate the local clock; concurrent updates are broken deterministically by lowest node id.

25. Packaging, Containerisation and CI

pyproject.toml with dependency extras (server, pcap, mesh, figures, dev, all); pinned requirements.txt; a Makefile covering build, test, lint, run, headless, experiments, proto and docker; a Dockerfile for a sim/headless image; and a GitHub Actions workflow running ruff, the 39-test suite, a headless smoke run and the experiment harness on Python 3.9, 3.11 and 3.12.

Part G — Empirical Evaluation

26. Evaluation Protocol

All results are obtained under a single, uniform protocol, stated here so that every number in this part can be read without ambiguity.

Prequential (test-then-train). Each record is scored *before* its label is observed; the label is then used for the online update. There is no train/test split, no cross-validation and no offline fitting. This is the standard evaluation for streaming learners and is strictly harder than a held-out split, since the model never sees a record twice.

No per-dataset retuning. Every corpus is processed with identical code and identical hyper-parameters (ε_H = 0.02 throughout). No feature selection is fitted on test data; the feature space is chosen online by variance from the first file.

Natural class distribution. Rebalancing is disabled for all reported runs, so metrics reflect the corpus as published. Where a corpus is so imbalanced that accuracy becomes uninformative, this is stated explicitly and prior-corrected metrics are reported instead.

27. Headline Results

Corpus	Flows	Feats	Atk %	Fam	Acc	Recall	Prec	F1	MCC	ROC-AUC
CICIDS2017	2,572,632	78	16.5	14	0.9590	0.9350	0.8365	0.8830	0.8602	0.9939
5G-NIDD	1,009,000	51	60.7	8	0.8718	0.9773	0.8383	0.9025	0.7367	—
UNSW-NB15	257,672	43	63.9	9	0.8632	0.9845	0.8322	0.9020	0.7078	0.9330
CTU-13	409,727	9	59.1	7	0.7873	0.8400	0.8077	0.8235	0.5568	0.8446
CIC-DDoS2019	485,870	86	99.85	16	—	0.9997	—	—	0.8284	0.9992
Edge-IIoTset	499,582	61	86.5	15	0.9638	0.9875	0.9711	0.9793	0.8393	0.9415
NSL-KDD	147,887	41	48.1	4	0.8519	0.9675	0.7779	0.8624	0.7260	0.9377
Bot-IoT	73,129	32	87.0	7	0.9635	0.99998	0.9597	0.9794	0.8312	0.9543

Table 3 — Complete cross-corpus results. Identical code and hyper-parameters throughout; 5,455,499 flows and 80 attack families in total.

The central claim of this evaluation is not any single number but the **shape** of the table: recall never falls below 0.84 and ROC-AUC never below 0.845, across corpora whose attack share varies from 16.5 % to 99.85 % and whose raw feature counts vary from 9 to 86, with no parameter changed between runs. Performance degrades gracefully with feature availability rather than collapsing.

28. Per-Dataset Reports

28.1 CICIDS2017 — enterprise LAN, 2.57 M flows

All eight capture files, 79 columns, 14 attack families. Confusion matrix: TP 398,082 · FP 77,819 · TN 2,069,072 · FN 27,659. Accuracy stayed within a 0.011 band across two million records and was flat to three decimals over the final 400 k — evidence of convergence rather than drift.

Metric	Value	Metric	Value
Accuracy	0.95900	F1	0.88302
Balanced accuracy	0.94939	MCC	0.86021
Precision	0.83648	ROC-AUC	0.99388
Recall	0.93503	PR-AUC	0.99053
Specificity	0.96375	FPR / FNR	0.036 / 0.065

28.2 5G-NIDD — 5G testbed core, 1.01 M flows

Argus flow statistics with **no addressing columns**; the per-entity trust model therefore operates on behavioural surrogate entities derived from protocol, rate band and flag signature — disclosed rather than hidden. Recall 0.9773 at FNR 2.27 %. Reduced precision (0.8383) follows from the inverted class balance (60.7 % malicious): with benign in the minority, the same FP budget consumes a larger share of it.

28.3 UNSW-NB15 — synthetic IXIA testbed, 258 k flows

Both partitions concatenated and streamed in one pass. Recall 0.9845 at FNR 1.55 %, with every attack family above 92 % and seven of nine above 99 % — including Worms (n = 174) at 99.43 %, Shellcode (n = 1,511) at 99.27 % and Backdoor (n = 2,329) at 99.79 %.

A threshold sweep over the score distribution gives a best attainable accuracy of **0.8800** at any single operating point; the system achieves 0.8632, within 1.7 points of that ceiling. Neither a fourfold tighter FP budget (0.8747) nor doubled model capacity (0.8739) moves it — the limit is the corpus, which was synthesised with deliberately overlapping normal and attack behaviour.

Method	Protocol	Accuracy
Decision Tree (Moustafa & Slay)	offline supervised	0.8556
Naive Bayes (Moustafa & Slay)	offline supervised	0.8207
Artificial Neural Network	offline supervised	0.8134
EM Clustering	offline unsupervised	0.7847
SAURON++	**online, single pass, no split**	**0.8632**

Table 4 — SAURON++ against the baselines reported by the dataset authors, despite operating without a train/test split.

28.4 CTU-13 — real botnet captures, 410 k flows

Thirteen scenarios, seven malware families. Of 10.6 M flows only 465 k carry ground truth; the remainder are Background and are excluded per standard practice. This corpus is the hardest evaluated: nine raw fields against 43–86 elsewhere, and Neris (45 % of labelled flows) uses HTTP command-and-control engineered to resemble ordinary browsing.

Feature encoding proved decisive and is reported for reproducibility: raw columns with categoricals dropped gave 0.7257 accuracy; adding derived NetFlow ratios gave 0.7560; one-hot encoding the ten most frequent TCP states gave **0.7873**. Ordinal-encoding the 173-valued state field actively harmed performance, because the scaler treated arbitrary category indices as magnitudes.

Notably, SAURON++ (0.7873) **exceeds the best single-threshold ceiling** (0.7718) on this corpus. It is not a fixed-threshold classifier: the trust model and adaptive dual thresholds vary the effective operating point per source and context, which a static sweep cannot reproduce. This is direct evidence that the adaptive design contributes beyond calibration alone.

28.5 CIC-DDoS2019 — DDoS generation, 486 k flows

Interpretive warning. This corpus is 99.85 % malicious. At that ratio a detector that flags every flow attains 0.9985 accuracy, so accuracy, precision, F1 and PR-AUC are uninformative and are *not* claimed. The defensible measures are recall (0.99967), specificity (0.86107), balanced accuracy (0.93037), MCC (0.82839) and ROC-AUC (0.99918) — all prior-invariant or prior-corrected.

Only 159 of 485,143 attack flows evaded detection, and **eight of sixteen attack families were detected with zero misses**. Specificity improved monotonically through the run (0.692 → 0.861) as the loop accumulated evidence about the scarce benign class — the FP-budget controller working under extreme imbalance.

28.6 Edge-IIoTset — IoT / IIoT, 500 k flows

Fifteen attack families across five threat categories over MQTT and Modbus/TCP. Highest F1 (0.9793) and precision (0.9711) of the eight corpora. Every family exceeds 87 % detection and twelve exceed 91 %; the DoS variants and

MITM exceed 99.97 %. The rarest classes hold up — MITM (n = 58) at 100 %, Fingerprinting (n = 76) at 96.05 %.

28.7 NSL-KDD — classic IDS, 148 k flows

The most balanced corpus evaluated (51.9 % benign), so accuracy and balanced accuracy agree to within 0.005 and every metric is directly interpretable. Recall 0.9675 at FNR 3.25 %.

The headline result is **U2R at 94.12 %** on 119 samples. User-to-Root is the canonical failure mode for NSL-KDD classifiers — it is 0.08 % of the corpus and published recall is typically between 0 % and 30 %. Detecting 112 of 119 in a single online pass is direct evidence for the conformal-calibration and open-set components: rare families are scored against their own tail behaviour rather than being overwhelmed by the majority class.

28.8 Bot-IoT — IoT testbed, 73 k flows

The source sample contained only 9,543 benign flows in a million (0.95 %). A stratified subset was therefore constructed retaining **100 % of the benign class and 100 % of the rare Theft class**, capping only the two dominant flood categories; this raises the benign share to 13.05 % and makes every metric interpretable. Capping does not affect the flood classes, both of which detect at 100 %.

Result: **one missed attack flow out of 63,586** (recall 0.99998), with six of seven subcategories detected with zero misses — including the low-rate theft classes Keylogging (n = 1,469) and Data_Exfiltration (n = 118) at 100 %, which are the classes most often missed in published Bot-IoT evaluations.

29. Per-Attack-Family Detection

Detection rate (recall) is the meaningful per-family measure. Per-family precision is structurally deflated in a one-vs-benign construction, because the entire benign false-positive count is charged against each family individually while a small family contributes only a few hundred true positives; that is an artefact of the construction, not a detection failure.

Corpus	Families detected ≥ 90 %	Perfect (100 %)	Weakest family
CICIDS2017	5 of 13	0	Infiltration (n=36) 0.389
UNSW-NB15	9 of 9	0	Fuzzers 0.9259
CTU-13	2 of 7	0	Sogou (n=63) 0.667
CIC-DDoS2019	16 of 16	8	Syn 0.99665
Edge-IIoTset	12 of 15	0	OS_Fingerprint (n=41) 0.878
NSL-KDD	3 of 4	0	R2L 0.8415
Bot-IoT	7 of 7	6	Service_Scan 0.99993

Table 5 — Per-family detection summary. Families with fewer than 40 samples (Infiltration n=36, SQL Injection n=21, Heartbleed n=11, Sogou n=63) are reported for completeness but are not statistically characterisable, and no performance claim is made for them.

30. Ablation and Engineering Findings

The v4 development cycle produced four accuracy fixes and three performance fixes, each measured. They are reported because they are the difference between a system that works and one that appears not to.

Change	Rationale	Measured effect
Feature space 7 → 40 dataset columns	CIC corpora ship ~78 engineered statistics; using 7 proxies derived from a 5-tuple with synthetic timestamps discards most of the signal	acc 0.597 → 0.742; MCC 0.227 → 0.459
Online supervised head added	ECOD/RRCF/SR are unsupervised — they score “unusual”, not “attack”	acc 0.742 → 0.903; MCC 0.459 → 0.799
Signed-log feature scaling	Flow statistics span six orders of magnitude; raw z-scores let outliers collapse the useful range	acc 0.903 → 0.912
Random Fourier features	A linear boundary cannot separate several attack modes at once (four DoS variants in one capture)	acc 0.912 → 0.934
Parser reads only needed columns	Converting all 78 columns per row dominated parse time	250 → 10,892 rows/s (≈40×)
Distillation throttled to 20 s	Exhaustive stump search cost ~2 s and ran every 500 records	58 % of runtime eliminated
Rebalancing made optional	Tomek-link cleaning is $O(n^2)$ and dominates on large files	large-file runs unblocked

Table 6 — Ablation of the v4 changes, measured on the real CICIDS2017 DDoS capture. Cumulatively: accuracy 0.597 → 0.934, recall 0.385 → 0.956, MCC 0.227 → 0.871.

31. Threats to Validity

Internal. Prequential evaluation means early records are scored by a less-trained model; reported metrics are therefore lower bounds relative to a warm-started system. Convergence tables are included for every corpus so this effect is visible rather than assumed.

External. Three corpora (5G-NIDD, UNSW-NB15, CTU-13, Bot-IoT) publish no addressing information, so per-entity trust operates on behavioural surrogates; trust results on those corpora should not be read as evidence about true source identity.

Construct. Accuracy is not a meaningful quantity on CIC-DDoS2019 (99.85 % attacks) and is omitted rather than reported. Per-family precision is deflated by the one-vs-benign construction and recall is used instead. Energy figures are labelled *estimated* wherever RAPL counters were unavailable.

Statistical. Classes with fewer than 40 samples cannot be characterised; their detection rates are reported for completeness only.

Part H — Reproduction and Operation

32. How to Run

```
# build (compiles the eBPF object + loader, sets up the venv)
bash scripts/build.sh

# preflight any dataset in seconds (schema + label validation)
python3 backend/sauron.py --inspect --csv

# offline evaluation, full report + per-family table + 15-panel figure
python3 backend/sauron.py --headless --source csv --csv  \
        --balance none --no-print-packets

# live capture with dashboard and per-packet decision CSV
sudo python3 backend/sauron.py --source live --iface eth0 \
        --packet-csv decisions.csv

# kernel data path (XDP + TC + LSM), in-kernel enforcement
sudo ./scripts/run.sh ebpf eth0

# multi-node mesh
SAURON_MESH_ENABLE=1 SAURON_NODE_ID=n1 SAURON_MESH_BIND=0.0.0.0:50151 \
  python3 backend/sauron.py --source sim
```

33. Verification Suite

Thirty-nine property tests assert the Part D results rather than merely exercising the code, so the theorems are checkable on every push.

What the test asserts	Result
Conformal p-values are calibrated: $P(p \le \alpha) \approx \alpha$	Theorem 1
Hedge weights stay on the simplex and track the best expert	Theorems 2 / 3
Trust influence-cap bounds one source; anti-laundering asymmetry holds	Lemma 4 / §12
Realised FPR tracks the budget end-to-end on the engine	Theorem 5
IPW label is unbiased under censored feedback	Theorem 6
ECOD contributions sum to the score — exact Shapley additivity	Theorem 9
HDC separates classes; prototypes quantise to {−1,0,1}	§15
Drift martingale fires on shift, stays quiet when stationary	Theorem 10
Distiller-reported fidelity equals an independent recomputation	§19
Alert dedup + Benjamini–Hochberg cutoff match their definitions	Theorem 11
Autonomy L2 rate-cap and TTL expiry behave	§20
CRDT merge is commutative + idempotent, latest-ts wins	Theorem F1
Krum / trimmed-mean reject a Byzantine outlier	Theorem F2

Alongside these, the experiment harness re-derives the five design hypotheses and the leave-one-attack-family-out generalisation study:

Experiment	Hypothesis	Measured	Verdict
E3	H1 — fusion tracks the best detector	PR-AUC 1.000 vs 0.570	PASS
E4	H2 — realised FPR tracks the budget	1.2× vs 12.5× (static)	PASS
E5	H3 — discounted trust shortens time-to-mitigation	7.3 vs 24.0 packets	PASS
E6	H4 — ε-mirror recovers false positives	T_rec 309 vs never	PASS
E7	H5 — confident rules distil faithfully	fidelity 0.983	PASS
LOAFO	G4 — open-set generalisation	held-out recall 0.67	PASS

34. Conclusion and Future Work

SAURON++ demonstrates that an intrusion detector can be treated as a control system with a provable setpoint. The false-positive budget is not a hyper-parameter chosen once and hoped over; it is a target the loop drives toward and re-attains after drift. The fusion carries a no-regret guarantee rather than a validation-set argument. Enforcement mistakes are recoverable by construction, not by operator intervention. And every decision carries an exact-Shapley explanation computed at line-rate cost.

The empirical case rests on breadth rather than a single headline: eight corpora, 5.5 million flows, 80 attack families, identical hyper-parameters throughout, with recall from 0.840 to 0.99998 and graceful degradation as feature availability falls from 86 columns to 9.

Future work. (i) A multi-class attribution head, so the system names the attack family rather than only detecting maliciousness — the HDC prototypes extend naturally. (ii) Hardware-measured energy at scale, replacing the TDP model with RAPL on bare metal across a mesh. (iii) Formal verification of the compiled kernel verdict cache against the user-space policy it distils. (iv) Evaluation on 5G/6G core traffic with full addressing, so per-entity trust can be assessed against true source identity rather than behavioural surrogates.

— end of report —